DATA VISUALISATION 2

Semester 2, 2026

Version history

- 6 September 2026: Version 1.0

Submission

The submission is due on Sunday of Week 12, 25 October 2026, 11:55 PM.

Introduction

Similar to the Data Visualisation 1 assignment, you will design and build an effective data visualisation for a specific domain of your choice. This task involves critically evaluating information in a domain, developing your own visualisation, and telling a story. The major difference for Data Visualisation 2 is that you will use the Vega-Lite library to create maps and diagrams.

The aim of the assignment is to apply the data visualisation techniques discussed throughout the semester and demonstrate their use in an innovative context.

The following criteria apply specifically to the Data Visualisation 2 assignment:

- You are required to use the Vega-Lite or Vega libraries for creating maps and diagrams. You may use other libraries for creating diagrams that are not feasible with Vega or Vega-Lite, however, you must obtain approval from your tutor before doing so.
- The final deliverable consists of a publicly accessible web page hosted on your GitHub account. The JSON description for each diagram or map must be easily accessible in the same GitHub repository and formatted for human readability.
- You may use the Pure.css library or any other JavaScript/CSS/HTML library, authoring software or AI tools to create your web page.
- Your visualisation must load quickly, meaning that the total size of downloadable data should be less than a few megabytes. If your data is larger, you need to find ways to reduce the size.
- Your web page must include at least three map idioms.

The following criteria are copied from the Data Visualisation 1 assignment and also apply for this assignment:

- Why? The visualization addresses a particular need within a specific domain.
- What? It visualises data relevant to the domain.

- Who? Design your visualisation for the average Australian or Malaysian, depending on whether you are a student at the Clayton or Malaysia campus.
- The visualisation must transform data into meaningful information and provide insight that would otherwise be difficult or impossible to obtain.
- It must provide an appropriate level of interactive exploration.
- It needs to show some innovation. While it does not have to be wholly original, it must not be a replica of an existing visualisation.
- The submitted visualisation should adhere to the submitted sketch.
- Your submission must demonstrate the use of Munzer's What/Why/How framework.
- It must apply design principles discussed throughout the unit, such as data-ink ratio, storytelling, layout, typography and visualisation idioms with appropriate use of marks and channels.

Task Description and Schedule

1. During the Week 1 studio, your tutor will assign you a domain and a specific topic.
2. Search relevant data that is publicly available.
3. During the studio of Week 7, sketch your visualisation on an A4 sheet. Discuss your sketch with your tutor in the Week 7 studio to obtain feedback.
4. Create the visualisation: Weeks 9 to 11: Create a web page and embed your diagrams and maps. Use CSS to style typography and design the layout. Make your visualisation publicly accessible as a GitHub page.
5. Using the template provided, submit the following on Moodle:
    a. The URL to your visualisation on your GitHub account. (URL to webpage)
    b. A URL to a PDF document with an image of your sketch on your GitHub account.
    c. A description of the following via a form on the Moodle submission page:
        i. The domain, the why and the who of the visualisation.
        ii. What: The data (sources, authors, relevance, creation process, etc.).
        iii. How: Give a rationale for choosing the specific idioms and explain how they help the users to achieve their tasks. Describe any special features of your visualisation, such as custom-built elements.
6. Attend the interview during the studio of Week 12 with questions about your visualisation and/or materials from Weeks 1 to 11.

Requirements

Note the following is a verbatim copy from the Data Visualisation 1 assignment, except for the first paragraph about maps and a few references to Tableau.

Maps: Include at least three different map idioms. The maps must use appropriate projections and idioms.

Data: The visualisation must use accurate and reliable real-world data; it must not use fabricated or tampered data. The most recent available dataset should be used. **You must**

combine data from two different sources. Use existing and reasonably clean data; creating datasets or performing data wrangling are not topics covered in this unit and will not result in a higher mark.

Format: The entire visualisation must be accessible from a single URL on your GitHub account. It must be viewable on a single web page that can be scrolled, and the full width of the visualisation must be visible on a small laptop screen without horizontal scrolling. There should be no buttons (or other web links) that swap major sections of the web page, though you may use buttons to show and hide visualisation elements.

Presentation not exploration: The goal of this assignment is to create a visualisation that communicates interesting information in an easily accessible and graphically engaging way using storytelling elements, layout principles, typography, and graphical design. The goal is not to create an expert tool for exploring and analysing a dataset.

Audience: The topic must be relevant to and easily understandable by an audience in Malaysia (for students in Malaysia) or Australia (for students in Australia). Avoid jargon and technical terms when possible. If specialised terminology is necessary, introduce and explain these terms. Avoid visualisations that require knowledge of statistics (e.g., understanding standard deviation or confidence intervals), as the general audience is not expected to have such expertise. If statistical concepts are necessary, introduce and explain them.

Number and complexity of visualisations: Your submission must contain a minimum of ten charts. A small-multiples chart, or a series of similar charts, count as a single chart. We are looking for carefully designed and annotated charts. For a C, D or HD grade, you need to create advanced idioms that are effective and meaningful. Advanced idioms are different to bar charts, stacked bar charts, grouped bar charts, dot plots, single- or multi-line charts, area charts, pie charts, scatterplots, and bubble charts.

Storytelling: The charts combined with text, icons, and pictures should guide the viewer through an interesting story using layout principles, typography, and graphic design principles. Your story needs to be structured in a logical sequence. Avoid pixelated images, unrelated graphical elements, or trivial text. Complement your visualisation with concise, informative, and grammatically correct narrative text that tells a story.

Interactivity: Integrate interactivity where it makes sense, but avoid adding interactive elements merely for their own sake.

Maps: Geographic maps will be required for the second visualisation assignment. For this first visualisation assignment, it is recommended to use non-geographic idioms; **geographic maps will be disregarded when marking your submission**.

Copyright: You are encouraged to use icons and other simple graphical elements where appropriate. As your visualisation will be publicly accessible, it is your responsibility to ensure that you have the right to use data, icons, images, etc. Consult with your tutor if in doubt. You need to indicate the source and URL (if available online) for any external elements you use, such as datasets, photos, articles, academic papers, and other sources

of information. There is no need to indicate the source and URL for simple icons, such as emojis, flags, or company logos. The use of AI tools for creating images, text or other elements, must be acknowledged on the submission.

Authorship: As your visualisation will be publicly accessible, you need to indicate your authorship and date your work. You may choose to release your work under a copyleft license, but this is not a requirement and your choice of license does not influence your mark.

Plagiarism: We will follow up on any kind of academic misconduct. For this assignment, you cannot integrate graphics (such as diagrams, charts, information graphics, etc.) created by others. The submission must be your own genuine work. Simply plugging a dataset into an existing visualisation and making tweaks to layout, typography, colour design, etc. is considered plagiarism.

Use of Artificial Intelligence

Generative artificial intelligence (AI) tools can be used for this assignment. Any use of generative AI must be appropriately acknowledged (see Learn HQ). Since grammar and style of text are part of the assessment, you are encouraged to use AI tools to ensure that the text included in the visualisation is grammatically correct and stylistically appropriate. You can also use AI tools to create illustrative figures, get feedback for your layout or any other creative use that helps you build a better visualisation.

Marking

This assignment is worth 25% of the final unit mark. A detailed marking rubric is provided on the next page. The mark will be 0 if the web page or any embedded elements are not publicly accessible on your GitHub repository.

Only maps and diagrams created with Vega-Lite or Vega will be marked unless you have received prior approval from your tutor for an exception. If less than three maps are included, the maximum possible mark for "Visualisation: Idioms and Complexity" will be reduced from 10% to 5%. If your submission does not include a sketch or your sketch is for a different topic, a penalty of -3% is applied.

The late submission penalty is 5% per day and a cut-off of seven calendar days applies.

	HD (80–100)	D (70–79)	C (60–69)	P (50–59)	N (0–49)
Sketch 2% 0 if created with digital tools	Four or more clear sections. Headings, text, figures, with detailed, creative and varied charts and maps. Clean and tidy.	Sections not clear, with uneven length, or other issues. Headings, text, figures, with some details, standard charts and maps. Clean and tidy.	Sections are not logically structured. Headings, text, figures,without details, standard charts and maps.	Sections are not logically structured. Headings, text, figures,without details, standard charts and maps with limited variety.	Incomplete submission, repetition in sketches, outcome difficult to implement.
Idioms and complexity 8% Maximum 4% if less than three map idioms	At least 8 advanced idioms* across more than 10 charts†. All idioms are applied effectively and meaningfully, demonstrating a high-level of understanding of visual marks, channels and idioms.	At least 8 idioms, including 5 to 8 advanced idioms*, across more than 10 charts†. All idioms are applied effectively and meaningfully, demonstrating a strong understanding of visual marks, channels and idioms.	At least 8 idioms, including 2 to 4 advanced idioms*, across more than 10 charts†. All idioms are effective and meaningful. Visual marks and channels are applied correctly.	5 standard idioms (e.g., bar chart, line graph) and 10 or more charts†. Minor issue with use of visual marks and channels.	Inappropriate idioms, less than 5 standard idioms, or less than 10 charts†, or incomplete visualisation. Visual marks, channels or idioms not applied correctly.
	* **Basic idioms** include bar charts, stacked bar charts, grouped bar charts, dot plots, single- or multi-line charts, area charts, pie charts, scatterplots, and bubble charts. Other idioms are considered **advanced**. † A small-multiples chart, or a series of similar charts, count as a single chart.				
Layout, colour, figure-ground 3%	Balanced and symmetric layout clearly structured in columns/rows with good use of white space. All elements aligned with sight lines. Very clear visual hierarchy by using consistent colour and figure-ground.	Balanced and symmetric layout mostly structured in columns/rows with use of white space. Most elements aligned with sight lines. Visual hierarchy by using colour and figure-ground.	Somewhat balanced and symmetric layout not consistently structured in columns/rows. Arbitrary use of white space. Some elements aligned with sight lines. Some visual hierarchy.	Layout not balanced or not symmetric, no apparent layout structure. Some elements aligned with sight lines. Limited visual hierarchy.	Layout not balanced and not symmetric, no apparent layout structure. Most elements are randomly placed. No visual hierarchy by using colour and figure-ground.
Typography 2%	Advanced typography, (non-standard typeface matching the topic, excellent readability of main text, appropriate line height, size, weight, colour, line length, alignment, spacing of text elements, etc.)	Consistent typography (standard typeface and attention to typography).	Standard typography with minor issues, but still easy to read.	Typography with some issues that impact readability of text (e.g., overuse of highlighting, poor or inconsistent spacing of text, centred text blocks, long text lines, etc.)	Inconsistency in fonts, sizes and weights. Typography with poor readability (e.g., inappropriate typeface, poor or inconsistent font size, weight, line lengths, etc.)
Storytelling, annotations, grammar and style 5%	• Clearly structured and engaging storytelling guides the reader through the visualisation. • Extensive use of informative annotations on diagrams and text outside of diagrams. • Correct grammar. • Writing is easy to follow and understand. • Metadata (data sources, author, date, etc.) is complete and well formatted.	• Storytelling guides the reader through the visualisation. • Use of informative annotations on diagrams and text outside of diagrams. • Correct grammar. • Writing is mostly easy to follow and understand. • Metadata is complete and well formatted.	• Some guidance of the reader through the visualisation. • Use of coherent annotations on diagrams and text outside of diagrams. • Some grammar issues. • Writing is not always easy to follow and understand, jargon or obscure abbreviations are used. • Metadata is complete and well formatted.	• Limited guidance of the reader through the visualisation. • Limited amount of annotations on diagrams and text outside of diagrams, with some incoherence. • Grammar issues throughout. • Writing is not easy to follow and understand, jargon or obscure abbreviations are used. • Some Metadata is missing or not well formatted.	• No apparent guidance of the reader through the visualisation. • Small number of incoherent annotations on diagrams and text outside of diagrams. • Major grammar issues throughout. • Writing is difficult to follow and understand, jargon or obscure abbreviations are used. • Metadata is incomplete or poorly formatted.
Description of domain, who, what, why, how 2%	Domain, who, what, why, how are appropriately and succinctly described.	Domain, who, what, why, how are described with minor issues.	Domain, who, what, why, how are described with major issues.	Domain, who, what, why, how are partially described.	Substantial aspects of domain, who, what (e.g., data source), why, how are not described.
Interview 3%	Correct answers to all questions.	Minor issue in answers.	Two minor issues in answers.	Vague or evading answers, minor issues.	Multiple issues in answers.